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SAUDI DIGITAL ACADEMY

multi-agent story building system

Project Documentation

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Abstract

The Multi-Agent Creative Story Building System is an AI-powered application that generates creative stories from a simple user idea. Instead of using a single AI model to perform all tasks, the system uses multiple specialized agents that work together. Each agent has a specific role, such as planning the story, creating characters, building the world, and writing scenes.

The system is developed using Python, LangChain, LangGraph, OpenAI API, and Streamlit. Shared memory is used to help the agents exchange information and maintain consistency throughout the story.

The main goal of this project is to demonstrate how Multi-Agent systems can improve story generation by dividing complex tasks into smaller responsibilities and allowing agents to collaborate efficiently.

Introduction

Artificial Intelligence (AI) has become an important technology in many fields, including education, healthcare, business, and entertainment. One of the most interesting applications of AI is text generation using Large Language Models (LLMs).

Although LLMs can generate stories from simple prompts, they sometimes produce inconsistent characters, weak plots, or contradictory events in long stories. This happens because a single model is responsible for handling every storytelling task.

To solve this problem, this project introduces a Multi-Agent Creative Story Building System. Instead of using one AI agent, the system uses multiple specialized agents that work together to generate a complete and consistent story. Each agent has its own responsibility, making the storytelling process more organized and efficient.
The application provides a simple web interface where users can enter a story idea and receive a complete story generated through collaboration between different AI agents.

Problem Statement

Generating long and coherent stories using a single AI model can be difficult. Characters may change their personalities, story events may become inconsistent, and the plot may lose its direction.
Another challenge is that one AI model must perform many different tasks at the same time, including planning the story, creating characters, designing the world, and writing scenes.
This project addresses these problems by dividing the storytelling process into multiple specialized AI agents that work together while sharing information. This approach improves story quality, consistency, and organization.

Project Objectives

The objectives of this project are:

- Develop a Multi-Agent storytelling system.
- Generate stories from a simple user prompt.
- Assign different storytelling tasks to specialized AI agents.
- Improve story consistency using shared memory.
- Build an easy-to-use web interface using Streamlit.
- Demonstrate the use of LangGraph for coordinating multiple AI agents.

Project Scope

This project focuses on generating creative stories using multiple AI agents. The system accepts a story idea from the user and automatically produces:

- A story outline
- Character profiles
- World-building information
- Complete story scenes

The project does not include image generation, voice narration, or story editing by human users.

Technologies Used

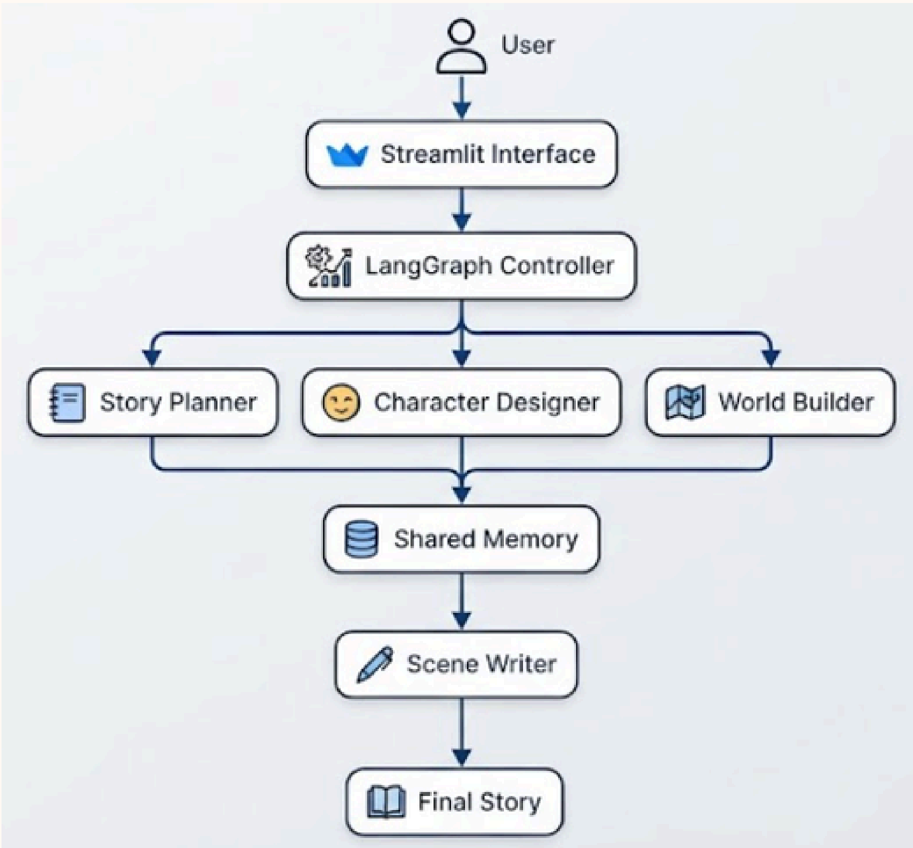
Several technologies were used to develop the Multi-Agent Creative Story Building System. Each technology has a specific role in the application.

Technology	Purpose
Python	Main programming language
LangChain	Connects the application with the language model and manages prompts
LangGraph	Controls the workflow between AI agents
OpenAI API	Generates story content using a Large Language Model
Streamlit	Creates the web interface
Shared Memory	Stores information shared between agents

System Architecture

The system follows a Multi-Agent architecture where several AI agents collaborate to generate a complete story. Instead of using one AI model for every task, each agent has a specific responsibility.

The process starts when the user enters a story idea. The idea is passed through several agents, and each one contributes to building the final story.



The user begins by entering a story idea through the Streamlit interface. The request is sent to LangGraph, which manages the execution order of the AI agents.

The Story Planner creates the main plot. Next, the Character Designer generates the characters based on the plot. Then, the World Builder creates the environment where the story takes place. Finally, the Scene Writer combines all previous outputs to produce the complete story.

Throughout the process, all agents use shared memory to access the information generated by previous agents. This helps maintain consistency between the plot, characters, and world.

Multi-Agent Design

The system is built around four specialized AI agents. Each agent performs one specific task, making the storytelling process more organized and efficient.

1. Story Planner
The Story Planner is the first agent in the workflow.
Responsibilities:
- Understand the user's idea.
- Create the story outline.
- Define the beginning, middle, and ending.
- Organize the main events.

Output:
- Story outline.

2. Character Designer
The Character Designer creates the characters that appear in the story.
Responsibilities:
- Generate the main characters.
- Define their personalities.
- Create relationships between characters.
- Add character backgrounds.

Output:
- Character profiles.

3. World Builder
The World Builder creates the environment where the story takes place.
Responsibilities:
- Design locations.
- Define the world's rules.
- Describe the environment.
- Create the story setting.

Output:
- World-building information.

4. Scene Writer
The Scene Writer is responsible for writing the final story.
Responsibilities:
- Write the narrative.
- Create dialogues.
- Connect events smoothly.
- Follow the outline created by previous agents.

Output:
- Complete story.

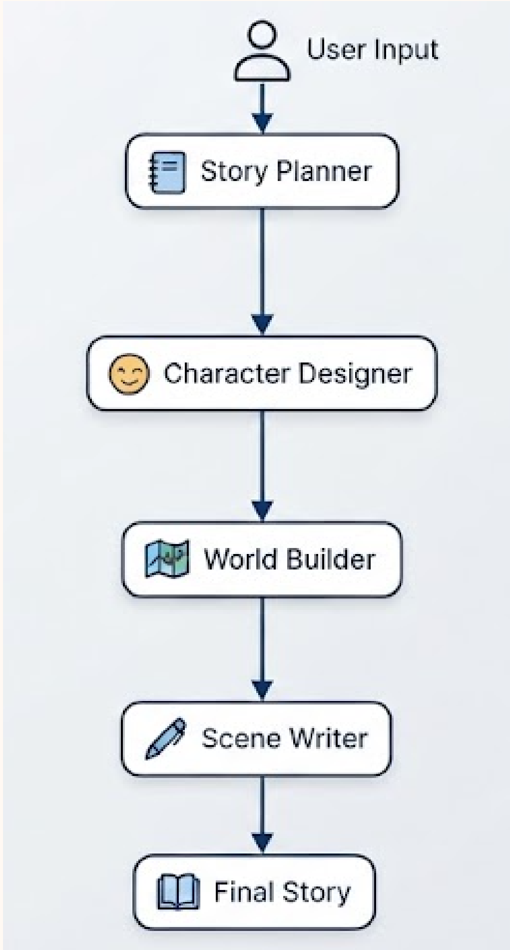
System Workflow

The Multi-Agent Creative Story Building System follows a step-by-step workflow where each AI agent performs a specific task before passing its output to the next agent.

The workflow begins when the user enters a story idea through the Streamlit interface. The Story Planner analyzes the idea and creates a structured outline. This outline is then sent to the Character Designer, which generates detailed character profiles.

Next, the World Builder creates the story environment, including locations, world rules, and background information. Finally, the Scene Writer combines all previous outputs to generate the complete story.

All generated information is stored in shared memory, allowing every agent to access the same context and maintain consistency throughout the storytelling process.



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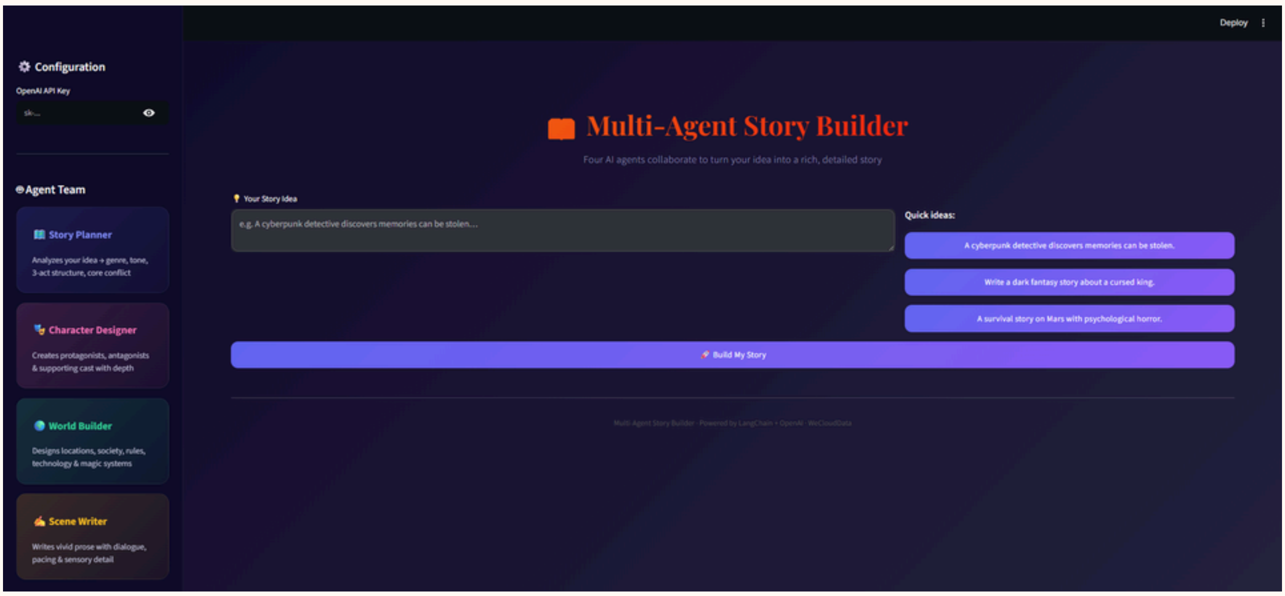

Shared Memory

Shared memory is one of the most important features of the system. Instead of allowing each AI agent to work independently, all agents can access the same stored information.

The shared memory contains:

- Story outline
- Character information
- World-building details
- Previous generated content

This approach helps the agents produce consistent outputs and reduces contradictions in the final story.

Benefits of Shared Memory

- Maintains character consistency.
- Prevents conflicting story events.
- Keeps the world rules consistent.
- Improves collaboration between agents.

Challenges

During the development of the project, several challenges were encountered.

API Dependency

The system depends on the OpenAI API, meaning that an internet connection and a valid API key are required.

Prompt Design

Designing effective prompts for each agent required multiple iterations to ensure that every agent focused only on its assigned responsibility.

Story Consistency

Maintaining consistency across long stories was challenging, which made shared memory an important part of the system.

Response Time

Because multiple agents work sequentially, generating a complete story takes longer than using a single AI model.

Future Work

Several improvements can be added to the system in future versions. Possible enhancements include:

- Add image generation for story illustrations.
- Support multiple languages.
- Add more specialized AI agents, such as a Dialogue Agent or Emotion Agent.
- Save generated stories in a database.
- Allow users to edit generated stories.
- Support local Large Language Models instead of cloud APIs.

Conclusion

The Multi-Agent Creative Story Building System demonstrates how collaboration between specialized AI agents can improve automated story generation.

By dividing the storytelling process into multiple stages, the system produces more structured and consistent stories than traditional single-agent approaches. The use of LangGraph allows the agents to communicate efficiently, while shared memory ensures that all generated information remains consistent throughout the story.

Overall, the project demonstrates the potential of Multi-Agent AI systems in creative writing and provides a strong foundation for future research and development in collaborative artificial intelligence applications.

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